

Credit Quantity, Credit Prices, and Future Output: A Model of Heterogeneous Financing Access

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Abstract

Preliminary evidence for Peru suggests that credit volume may contain more forecasting information for non-tradable activity at relatively short horizons, whereas a credit-price spread may become more informative for tradable activity over a longer horizon. This paper develops a deliberately small model that can rationalize that qualitative asymmetry. Firms invest at date zero and produce at date one. Bank credit is relatively cheap but capped; some firms can finance investment beyond the bank ceiling in an external market at a wedge. The firm's optimum has a quantity regime, in which the bank ceiling is binding, and a price regime, in which external finance funds the marginal unit. In the quantity regime, a larger credit ceiling raises investment and future output, whereas a small spread change has no local real effect. In the price regime, a larger bank ceiling changes financing composition but not investment, while a spread increase lowers investment and output. Aggregating firms shows that a sector with more bank-dependent firms is more quantity-sensitive and less price-sensitive. The model provides a theoretical bridge from financing heterogeneity to the sector-specific patterns being studied in the thesis. A quadratic specialization yields closed-form first-order conditions. A direct Lean 4 prototype checks the core algebra; the course-required AppliedModelingLib formalization remains part of the final-paper workflow.

Keywords: credit constraints; external finance; sectoral heterogeneity; investment; forecasting.

JEL codes: E22, E32, E44, G21, G32.

1 Introduction

Credit conditions can affect firms through two conceptually different margins. A quantity restriction limits how much a firm can invest even when a project has positive marginal value. A price change alters the marginal cost of funds and therefore the desired scale of investment. Treating these margins as interchangeable can hide economically meaningful heterogeneity.

Fernandez and Chavez (2026) compare autoregressive benchmarks with models augmented by credit indicators for Peruvian tradable and non-tradable sectors. Their work in progress points to a qualitative asymmetry: credit volume appears more informative for non-tradable activity at relatively short horizons, whereas a credit-price spread appears more informative for tradable activity over a longer horizon. These results remain preliminary and should not be read as a settled ranking of exact forecast horizons. Moreover, predictive content alone does not supply an optimizing model that explains why the relevant credit margin might differ across sectors.

This paper supplies that missing mechanism. Firms have internal funds and a bank credit line. Bank credit is cheaper than external market finance, but the line is capped. Firms differ in

whether, and at what wedge, they can use an alternative source after exhausting the bank line. Current financing choices fund capital that produces next-period output, so the model generates a direct lead-lag relation from credit conditions to real activity.

The main result is a regime separation. When the firm’s desired capital stock lies at the bank ceiling, quantity is locally decisive: relaxing the ceiling raises capital one for one, while a small funding-price movement leaves capital unchanged. When the firm uses market finance at the margin, price is decisive: the common spread changes desired capital, while a larger bank line merely replaces market finance with bank finance. Sectoral differences follow from composition. If non-tradables contain more bank-dependent firms than tradables, non-tradable output is more sensitive to credit quantity and tradable output is more sensitive to spreads.

The contribution is intentionally narrow. It is not a full financial accelerator or a structural interpretation of the thesis’s forecast tests. Instead, it isolates the minimum mechanism needed to convert the empirical price-versus-quantity asymmetry into a proposition with explicit conditions.

2 Related literature and positioning

The first strand studies equilibrium credit rationing. Stiglitz and Weiss (1981) show that asymmetric information can prevent the interest rate from clearing the loan market, so observationally similar borrowers may face a quantity restriction. Petersen and Rajan (1994) find that lending relationships benefit small firms primarily through greater availability of funds rather than large price reductions. The quantity regime below takes a credit ceiling as a reduced-form representation of those frictions; it does not reproduce their screening or relationship formation models.

The second strand studies balance-sheet and collateral amplification. Kiyotaki and Moore (1997) make durable assets both productive inputs and collateral, generating persistent responses through endogenous credit limits. Bernanke, Gertler, and Gilchrist (1999) embed an external-finance premium and heterogeneous capital-market access in a dynamic macroeconomic framework. Holmstrom and Tirole (1997) link firm net worth and intermediary capital to the supply of loanable funds. The present paper treats net worth and the credit ceiling as predetermined in its baseline. Section 9 shows how an endogenous ceiling can recover an accelerator without obscuring the main regime result.

The third strand establishes firm heterogeneity. Gertler and Gilchrist (1994) document that small firms contract more sharply after monetary tightening, while Petersen and Rajan (1994) emphasize their dependence on institutional lenders. These findings motivate a higher constrained share among non-tradable firms. This is a composition assumption to be disciplined by data, not a theorem that every small firm is non-tradable or every tradable firm has market access.

The fourth strand concerns leading financial indicators. Gilchrist and Zakrajsek (2012) construct a credit-spread measure with predictive power for future activity, and Faust et al. (2013) find that credit spreads improve real-time activity forecasts. Stock and Watson (2003) caution that predictive performance can be unstable across countries and periods. For Peru, Lahura (2011) finds a stable long-run credit-output relationship, while Butron and Pozo (2025) document a positive relationship using sectoral and regional heterogeneity.

The closest paper for the sectoral motivation is Muller and Verner (2024). They show that credit allocation toward non-tradable rather than tradable sectors predicts different macroeconomic outcomes, discuss smaller and more constrained non-tradable firms, and report robustness involving bond issuance. That evidence means neither the tradable/non-tradable split nor financing heterogeneity can be claimed as novel here. Holmstrom and Tirole’s (1994) working-paper version and their published 1997 model also already contain capital thresholds associated with monitored

and direct finance. Gilchrist, Sim, and Zakrajsek (2014) subsequently model the joint dynamics of investment and credit spreads.

The narrower course-paper contribution is to combine a capped, relatively cheap bank tranche with a marginal outside-finance tranche and derive a specific comparative-static ranking: a sector with more firms at the cap is more sensitive to observed credit quantity, while a sector with more firms financing the marginal unit externally is more sensitive to a common funding price. The reviewed papers do not state that quantity-versus-price ranking in this form. This is a provisional positioning claim, not a claim that the broader ingredients or empirical sector pattern are new.

3 Environment

There are two dates, t and $t + 1$, and sectors $s \in \{N, T\}$, interpreted as non-tradable and tradable. A firm enters date t with internal funds $n > 0$. It chooses capital k , bank borrowing b , and external market finance m . Capital becomes productive at $t + 1$, making date- t credit conditions leading indicators of output.

3.1 Technology

The general model uses a differentiable production function $F : \mathbb{R}_+ \rightarrow \mathbb{R}_+$.

Assumption 1 (Technology). $F(0) = 0$, $F'(k) > 0$, and $F''(k) < 0$ on the economically relevant domain. Future revenue is $F(k)$ and its date- t value is $\beta F(k)$, where $\beta \in (0, 1]$.

Closed forms and the Lean verification use

$$F(k) = ak - \frac{c}{2}k^2, \quad a > 0, \quad c > 0, \quad (1)$$

restricted to $k < a/c$, where output is increasing. This quadratic form is a local second-order representation of any smooth concave technology and keeps all regime boundaries transparent.

3.2 Financing

The firm's sources satisfy

$$k = n + b + m, \quad 0 \leq b \leq \bar{b}, \quad m \geq 0. \quad (2)$$

Bank finance has gross marginal cost

$$R_B(z) = R_f + z, \quad (3)$$

where z is a common credit-spread factor. External market finance costs

$$R_M(z, \tau) = R_B(z) + \tau, \quad \tau \geq 0. \quad (4)$$

The access wedge τ includes issuance costs, information costs, and other barriers to non-bank funding. A bank-only firm is the limiting case in which $m = 0$ is imposed. A market-access firm can choose $m \geq 0$.

Assumption 2 (Funding order). $0 < R_B \leq R_M < \beta a$. The firm uses internal funds first, bank credit second, and market finance last. The strict upper bound guarantees a positive target capital stock in the quadratic specialization.

The assumption that z shifts both funding costs is important. Empirically, the observed corporate loan spread is interpreted as a proxy for a broader external-finance condition, not as an isolated price that leaves bond and foreign-credit costs fixed. Section 9 discusses this identification boundary.

4 The firm problem

The firm maximizes discounted profits:

$$\max_{k,b,m} \beta F(k) - R_B b - R_M m \quad \text{s.t.} \quad (2). \quad (5)$$

Let the capital stock attainable after exhausting the bank line be

$$\bar{k} \equiv n + \bar{b}. \quad (6)$$

Because bank finance is weakly cheaper, the reduced problem is continuous and piecewise:

$$\Pi(k) = \begin{cases} \beta F(k) - R_B(k - n), & n \leq k \leq \bar{k}, \\ \beta F(k) - R_B \bar{b} - R_M(k - \bar{k}), & k > \bar{k}. \end{cases} \quad (7)$$

The left and right first-order conditions are

$$\beta F'(k) = R_B \quad (\text{bank-finance margin}), \quad (8)$$

$$\beta F'(k) = R_M \quad (\text{market-finance margin}). \quad (9)$$

Let k_B and k_M solve (8) and (9). Strict concavity makes both targets unique. Since $R_M \geq R_B$, declining marginal product implies $k_M \leq k_B$.

Under (1),

$$k_B = \frac{\beta a - R_B}{\beta c}, \quad k_M = \frac{\beta a - R_M}{\beta c}, \quad k_M \leq k_B. \quad (10)$$

Lemma 1 (Optimal capital with market access). *Under Assumptions 1–2, the unique optimal capital stock is*

$$k^* = \begin{cases} n, & k_B \leq n, \\ k_B, & n < k_B < \bar{k}, \\ \bar{k}, & k_M \leq \bar{k} \leq k_B, \\ k_M, & \bar{k} < k_M. \end{cases} \quad (11)$$

For a bank-only firm, the last segment is infeasible and $k^* = \min\{\max\{n, k_B\}, \bar{k}\}$.

Proof. On $[n, \bar{k}]$, the derivative of profit is $\beta F'(k) - R_B$; on $(\bar{k}, \infty)$ it is $\beta F'(k) - R_M$. Both are strictly decreasing. If $k_B \leq n$, the left derivative is already non-positive at n . If $n < k_B < \bar{k}$, it crosses zero at k_B before the kink. If $k_M \leq \bar{k} \leq k_B$, the derivative is weakly positive immediately to the left of the kink and weakly negative immediately to its right, so the kink is optimal. Finally, if $\bar{k} < k_M$, profit continues increasing after the bank line is exhausted and reaches its maximum at k_M . Strict concavity gives uniqueness, with adjacent formulas agreeing at equality boundaries. $\square$

5 Quantity and price transmission

The two empirically relevant cases are the kink and market-finance regimes.

Proposition 1 (Quantity versus price). *Suppose all inequalities defining a regime are strict, so sufficiently small parameter changes do not cause a regime switch.*

(i) If $k_M < \bar{k} < k_B$, the bank ceiling binds and

$$k^* = \bar{k} = n + \bar{b}, \quad \frac{\partial k^*}{\partial \bar{b}} = 1, \quad \frac{\partial k^*}{\partial z} = 0. \quad (12)$$

Hence

$$\frac{\partial y_{t+1}}{\partial \bar{b}} = F'(\bar{k}) > 0, \quad \frac{\partial y_{t+1}}{\partial z} = 0. \quad (13)$$

(ii) If $\bar{k} < k_M$, the firm uses market finance at the margin and

$$k^* = k_M(z, \tau), \quad \frac{\partial k^*}{\partial \bar{b}} = 0, \quad \frac{\partial k^*}{\partial z} = \frac{1}{\beta F''(k_M)} < 0. \quad (14)$$

Consequently,

$$\frac{\partial y_{t+1}}{\partial \bar{b}} = 0, \quad \frac{\partial y_{t+1}}{\partial z} = \frac{F'(k_M)}{\beta F''(k_M)} < 0. \quad (15)$$

A larger bank line replaces market finance one for one until $m = 0$; it raises profits by reducing funding costs but does not change capital or output.

Proof. Part (i) follows from the third line of (11). Since strict inequalities preserve the regime locally, $k^* = n + \bar{b}$ is independent of z . Differentiating $F(k^*)$ gives (13).

For part (ii), the market-margin condition is $\beta F'(k_M) = R_M(z, \tau)$. Implicit differentiation with respect to z , using $\partial R_M / \partial z = 1$, yields $\beta F''(k_M) \partial k_M / \partial z = 1$. Because $F'' < 0$, the response is negative. While $m > 0$, a marginal increase in $\bar{b}$ increases bank borrowing and reduces market finance by the same amount, leaving total capital unchanged. Applying the chain rule to output gives (15). $\square$

For the quadratic technology, the price-regime comparative statics become especially transparent:

$$\frac{\partial k_M}{\partial z} = -\frac{1}{\beta c}, \quad \frac{\partial F(k_M)}{\partial z} = -\frac{R_M}{\beta^2 c} < 0. \quad (16)$$

Remark 1 (Local zero effects). The zero derivatives are regime statements, not universal claims. A sufficiently large price shock can move a constrained firm out of the kink, and a sufficiently large increase in the bank ceiling can eliminate market borrowing. The model therefore predicts threshold nonlinearities.

6 Sectoral aggregation and empirical mapping

Let $\lambda_s \in [0, 1]$ be the share of sector- s activity generated by firms in the quantity regime. For a transparent aggregation, suppose firms share technology and financing parameters conditional on regime. Sectoral output is

$$Y_{s,t+1} = \lambda_s F(n + \bar{b}_s) + (1 - \lambda_s) F(k_M(z_s, \tau_s)). \quad (17)$$

Assumption 3 (Sector composition). *The non-tradable sector contains a larger share of bank-dependent, quantity-constrained firms:*

$$0 \leq \lambda_T < \lambda_N \leq 1. \quad (18)$$

Proposition 2 (Sectoral sensitivity ranking). *Under Assumption 3 and common conditional primitives,*

$$\frac{\partial Y_N}{\partial \bar{b}} > \frac{\partial Y_T}{\partial \bar{b}} \geq 0, \quad (19)$$

$$\left| \frac{\partial Y_N}{\partial z} \right| < \left| \frac{\partial Y_T}{\partial z} \right|. \quad (20)$$

In the quadratic specialization,

$$\frac{\partial Y_s}{\partial \bar{b}} = \lambda_s [a - c(n + \bar{b})], \quad (21)$$

$$\left| \frac{\partial Y_s}{\partial z} \right| = (1 - \lambda_s) \frac{R_M}{\beta^2 c}. \quad (22)$$

Proof. Differentiate (17) and apply Proposition 1. The quantity response is the positive within-regime marginal product multiplied by λ_s . The absolute price response is the positive within-regime magnitude multiplied by $1 - \lambda_s$. Therefore $\lambda_N > \lambda_T$ establishes both strict rankings. Equations (21)–(22) substitute the quadratic derivatives. $\square$

This proposition offers a candidate interpretation for the thesis. If bank-dependent small firms are more prevalent in non-tradable activities, variation in available bank credit contains more information about their feasible investment and subsequent production. If tradable firms are more likely to access alternative sources, observed bank quantity can change through substitution without a corresponding change in total finance. A broad spread may then be more informative about their marginal funding condition.

The model delivers a structural sign restriction for a forecasting equation, not a causal interpretation of an ADL coefficient:

$$\Delta \log Y_{s,t+h} \approx \eta_s^B \Delta \log \bar{b}_{s,t} - \eta_s^z \Delta z_{s,t} + u_{s,t+h}, \quad (23)$$

where $\eta_N^B > \eta_T^B \geq 0$ and $\eta_T^z > \eta_N^z \geq 0$. These inequalities are theoretical sign and ranking restrictions, not estimates. The two-date model distinguishes a near-term quantity channel from a potentially more persistent price channel but does not identify exact forecast horizons. Doing so would require gestation lags, adjustment costs, or a dynamic capital stock and is an extension rather than a hidden assumption.

7 Transmission mechanism and qualitative predictions

The timing is central to the forecasting interpretation. At date t , the bank ceiling and the common funding factor are observed before capital becomes productive at $t + 1$. They can therefore predict future activity even when the model makes no causal claim about the reduced-form forecasting coefficient. The predictive content comes from different sufficient statistics in the two regimes.

Quantity-constrained firms. At the kink, the unconstrained bank-financed target lies above the cap, while the market-financed target lies below it. The positive left derivative of profits means that an additional unit of cheap bank funding would be installed immediately. Observed bank credit is therefore informative about the feasible investment pipeline, and the multiplier in (13) is the marginal product of the capital unlocked by the line. A small spread movement does not change capital while the firm remains at the kink; its first-order real effect is locally muted.

Market-access firms. Once outside finance funds the marginal unit, bank lending is no longer a sufficient statistic for total finance. A change in the bank line can induce a one-for-one substitution between bank and market funding with no change in capital. The common funding factor instead enters the marginal first-order condition directly. It changes the desired capital stock and, with installation or project-completion lags, can affect output over a more extended horizon. This mechanism requires the measured spread to proxy for the relevant common external-finance condition; it does not require literal bond issuance by every tradable firm.

Qualitative empirical implications. The model suggests three checks rather than a mechanical mapping from theory to forecast peaks. First, quantity predictive content should be stronger where bank dependence or the frequency of exhausted credit lines is higher. Second, spread predictive content should be stronger where firms have more alternative funding access and their marginal cost co-moves with the measured spread. Third, an expansion of bank lines should raise total investment for firms at the cap but mainly change financing composition for firms already using outside finance. These cross-sectional interactions are sharper tests of the mechanism than any single preferred forecast horizon.

8 Numerical illustration

Consider $\beta = 0.95$, $a = 2$, $c = 0.5$, $n = 0.4$, $\bar{b} = 0.5$, $R_B = 0.5$, and $R_M = 1.4$. Then

$$\bar{k} = 0.9, \quad k_B = 2.947, \quad k_M = 1.053. \quad (24)$$

A bank-only firm is quantity constrained at $k = 0.9$. A market-access firm uses 0.5 of bank credit and approximately 0.153 of market finance to reach $k = 1.053$.

Table 1: Illustrative regime responses

Experiment	Quantity-regime firm	Price-regime firm
Baseline capital	0.900	1.053
Baseline output	1.598	1.828
$\bar{b}$ rises by 0.10	capital rises to 1.000	capital unchanged
z rises by 0.02	capital unchanged	capital falls to 1.011
Financing adjustment	more total bank credit	bank/market substitution

With $\lambda_N = 0.8$ and $\lambda_T = 0.2$, the marginal quantity responses are 1.24 and 0.31, respectively. The absolute spread responses are about 0.62 and 2.48. The values are illustrative comparative statics, not a calibration to Peru.

9 Extensions and limitations

Endogenous bank ceiling. Let $\bar{b}_{i,t} = \phi_s n_{i,t}$, with pledgeability $\phi_s > 0$. In the quantity regime, $k_{i,t} = (1 + \phi_s)n_{i,t}$, so a net-worth shock changes investment directly and through borrowing capacity. Adding $n_{i,t+1} = \omega \Pi_{i,t}$ creates a simple financial accelerator. This extension is useful for propagation, but not necessary for the core price-versus-quantity proposition.

Continuous market access. The baseline binary distinction can be replaced by heterogeneous wedges τ_i or finite outside-finance limits $\bar{m}_i$. Lower wedges make $\bar{k} < k_M$ more likely; higher wedges make the kink regime more likely. The sectoral share λ_s can therefore be derived from a distribution of net worth, pledgeability, and access wedges.

Dynamic capital and horizons. A disciplined dynamic version introduces $K_{t+1} = (1-\delta)K_t + I_t$, a convex installation cost $\Phi_s(I_t)$, and the same two financing tranches for investment. Let $q_{i,t}$ denote the current-value shadow price of installed capital. The investment and capital first-order conditions imply

$$q_{i,t} = R_{j,t} + \Phi'_s(I_{i,t}) + \mu_{i,t}, \quad (25)$$

$$q_{i,t} = \beta \mathbb{E}_t [F'_s(K_{i,t+1}) + (1-\delta)q_{i,t+1}], \quad (26)$$

where $j = B$ when the marginal unit is bank financed and $j = M$ when it is market financed. The term $\mu_{i,t} \geq 0$ is the shadow wedge attached to a binding quantity limit; it is zero away from that limit (with the bank/market cost gap instead determining which tranche is marginal). Combining (25)–(26) yields the Euler equation

$$R_{j,t} + \Phi'_s(I_{i,t}) + \mu_{i,t} = \beta \mathbb{E}_t [F'_s(K_{i,t+1}) + (1-\delta)\{R_{j',t+1} + \Phi'_s(I_{i,t+1}) + \mu_{i,t+1}\}]. \quad (27)$$

Thus a credit-quantity innovation operates through the current and expected shadow wedges, whereas a spread innovation changes the marginal financing price directly. Sector-specific curvature in Φ_s , depreciation, or project gestation lags can shift the output response across horizons and may rationalize why preliminary quantity evidence appears at relatively short horizons while price information appears over a longer horizon. Estimating those dynamic parameters is left outside the current proposition so that the first deliverable remains exactly provable by hand; equation (27) states the disciplined route for the next version rather than claiming to identify the observed horizons already.

Measurement. The model's z is a common marginal-finance factor. The thesis uses a bank loan spread assigned by firm size because sector-level rates are unavailable. The empirical proxy may therefore contain measurement error and need not move one for one with bond or foreign-credit costs. A stronger empirical design would test whether the spread predicts substitution across funding sources.

Boundaries of the result. The model abstracts from default, risk choice, lender optimization, general equilibrium, and feedback from future expected output to current credit supply. It explains a conditional transmission mechanism, not the origin of the credit ceiling or the causal identification of credit shocks.

10 Conclusion

The same observed credit system can transmit shocks through different margins. For a bank-dependent firm at a binding ceiling, additional credit raises investment and future output. For a firm that can finance the marginal unit in an external market, a larger bank line changes the source of funds but not the scale of investment; the common funding spread governs that scale. A higher share of the first type makes non-tradable activity quantity-sensitive, while a higher share of the second makes tradable activity price-sensitive.

This mechanism turns the thesis’s preliminary forecasting pattern into a formal economic hypothesis with explicit assumptions, first-order conditions, regime boundaries, and falsifiable predictions. Its value for the course is not that it replaces the empirical thesis, but that it supplies the model the empirical exercise previously treated as a black box.

A Additional derivations

For the quadratic technology, profit on a segment with marginal funding cost R is, up to constants,

$$\pi_R(k) = \beta \left(ak - \frac{c}{2}k^2 \right) - Rk. \quad (28)$$

The FOC and second-order condition are

$$\pi'_R(k) = \beta(a - ck) - R = 0, \quad \pi''_R(k) = -\beta c < 0. \quad (29)$$

Completing the square gives the global identity

$$\pi_R(k_R) - \pi_R(k) = \frac{\beta c}{2}(k - k_R)^2 \geq 0, \quad (30)$$

where $k_R = (\beta a - R)/(\beta c)$. At the bank ceiling, global optimality in the kink regime follows from

$$\Pi(\bar{k}) - \Pi(k) = (\bar{k} - k)[\beta F'(\bar{k}) - R_B] + \frac{\beta c}{2}(\bar{k} - k)^2 \geq 0, \quad k \leq \bar{k}, \quad (31)$$

$$\Pi(\bar{k}) - \Pi(k) = (k - \bar{k})[R_M - \beta F'(\bar{k})] + \frac{\beta c}{2}(k - \bar{k})^2 \geq 0, \quad k \geq \bar{k}. \quad (32)$$

These are the identities formalized by the Lean theorem `kink_regime_optimal`.

B Cobb-Douglas robustness

If $F(k) = Ak^\alpha$ with $A > 0$ and $0 < \alpha < 1$, the two unconstrained targets are

$$k_j = \left(\frac{\beta \alpha A}{R_j} \right)^{\frac{1}{1-\alpha}}, \quad j \in \{B, M\}. \quad (33)$$

The policy function retains exactly the four cases in Lemma 1. In the market-finance regime,

$$\frac{\partial k_M}{\partial z} = -\frac{k_M}{(1-\alpha)R_M} < 0, \quad \frac{\partial F(k_M)}{\partial z} = -\frac{\alpha}{1-\alpha} \frac{F(k_M)}{R_M} < 0. \quad (34)$$

Thus the regime ranking is not an artifact of the quadratic technology.

C Lean formalization status

Table 2 records the declaration-level status of every numbered result. The current files in `lean/prototype/` are a direct Lean 4 prototype: they verify the main quadratic identities and sensitivity inequalities without `sorry`, but they are not yet the complete folder generated by the course’s AppliedModelingLib workflow. Before the final paper, the merged paper PDF will be pinned to a commit, the workflow will be run on that version, and the entire generated folder will replace the prototype.

Lean verifies mathematics conditional on supplied assumptions. It does not verify that those assumptions hold in Peruvian data, that the literature review is exhaustive, or that the empirical estimates identify causal effects. Paper commit formalized: **pending**; AppliedModelingLib **check-fast**: **pending**.

Table 2: Paper result to Lean declaration map

Paper result	Prototype declaration(s)	Current status
Lemma 1	<code>kink_regime_optimal; market_regime_optimal</code>	Core kink and market branches proved; full four-case wrapper pending Applied-ModelingLib run.
Proposition 1	<code>quantity_output_gain; market_target_decreases</code>	Proved for the quadratic specialization and strict local regimes.
Proposition 2	<code>quantity_sensitivity_order; price_sensitivity_order</code>	Proved under $\lambda_T < \lambda_N$ and the positivity assumptions stated in the paper.

D AI collaboration log

AI assistance was used to search and compare the literature, propose the two-tranche financing model, check symbolic and numerical claims, draft LaTeX, and prototype Lean statements. The author retained responsibility for the economic question, the interpretation of the thesis evidence, and each accepted assumption. Raw prompts and relevant answers are preserved in `prompts.md`.

Table 3: Claims checked during AI-assisted development

AI-assisted claim	Independent check	Verdict
A binding cheap-bank cap makes capital locally quantity-sensitive and price-insensitive. Outside finance makes the spread relevant and the bank line irrelevant for total capital.	Hand derivation, regime and inequalities, <code>code/verify.py</code> . FOC, implicit derivative, numerical check.	Accepted only for strict local kink regimes; large shocks can switch regimes. Accepted while market finance remains strictly positive; bank credit still affects profits and funding composition.
Sector composition generates the tradable/non-tradable ranking.	Direct differentiation of (17) and literature audit.	Accepted as a conditional prediction, not as proof that the composition assumption holds in Peru.
The model explains exact empirical forecast horizons.	Compared the static timing with the unfinished thesis results.	Rejected. The draft now states only a qualitative short-versus-longer-horizon interpretation.
The sector pattern itself is novel.	Checked Muller and Verner (2024), its appendices, and related working papers.	Rejected. Novelty is limited to the explicit quantity-versus-price comparative-static ranking.

References

- [1] Bernanke, B. S., Gertler, M., and Gilchrist, S. (1999). The financial accelerator in a quantitative business cycle framework. In J. B. Taylor and M. Woodford (eds.), *Handbook of Macroeconomics*, Vol. 1C, 1341–1393.
- [2] Butron, N., and Pozo, J. (2025). Dynamic relationship between credit and economic growth. BCRP Working Paper 2025-011.

- [3] Fernandez, S. A., and Chavez, A. M. (2026). *Credit Indicators and Sectoral Activity in Peru: Evidence on Predictive Power, 2008–2024*. Undergraduate research thesis, Universidad del Pacifico.
- [4] Faust, J., Gilchrist, S., Wright, J. H., and Zakrajsek, E. (2013). Credit spreads as predictors of real-time economic activity: A Bayesian model-averaging approach. *Review of Economics and Statistics*, 95(5), 1501–1519.
- [5] Gertler, M., and Gilchrist, S. (1994). Monetary policy, business cycles, and the behavior of small manufacturing firms. *Quarterly Journal of Economics*, 109(2), 309–340.
- [6] Gilchrist, S., and Zakrajsek, E. (2012). Credit spreads and business cycle fluctuations. *American Economic Review*, 102(4), 1692–1720.
- [7] Gilchrist, S., Sim, J. W., and Zakrajsek, E. (2014). Uncertainty, financial frictions, and investment dynamics. NBER Working Paper 20038.
- [8] Holmstrom, B., and Tirole, J. (1997). Financial intermediation, loanable funds, and the real sector. *Quarterly Journal of Economics*, 112(3), 663–691.
- [9] Kiyotaki, N., and Moore, J. (1997). Credit cycles. *Journal of Political Economy*, 105(2), 211–248.
- [10] Lahura, E. (2011). An empirical analysis of the credit-output relationship: Evidence from Peru. BCRP Working Paper 2011-018.
- [11] Muller, K., and Verner, E. (2024). Credit allocation and macroeconomic fluctuations. *Review of Economic Studies*, 91(6), 3645–3676.
- [12] Petersen, M. A., and Rajan, R. G. (1994). The benefits of lending relationships: Evidence from small business data. *Journal of Finance*, 49(1), 3–37.
- [13] Stiglitz, J. E., and Weiss, A. (1981). Credit rationing in markets with imperfect information. *American Economic Review*, 71(3), 393–410.
- [14] Stock, J. H., and Watson, M. W. (2003). Forecasting output and inflation: The role of asset prices. *Journal of Economic Literature*, 41(3), 788–829.